

Report for NoSQL Integration in a Medical Data Management System

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Use Case: *Medical Data Management System*

The project demonstrates the integration of three NoSQL database types—document (MongoDB), Graph (Neo4j), and Key-Value (Redis)—for efficient medical data management, focusing on patients, doctors, and appointment handling.

Implementation Overview:

1. Document Model (MongoDB):

MongoDB was used to store structured data, including:

- **Doctors:** Information such as name, specialty, and unique IDs.
- **Patients:** Name, age, medical history, and appointments.
- **Operations:** Insertion, querying, and updates for doctors, patients, and appointment records.

2. Graph Model (Neo4j):

Neo4j was leveraged to model and analyze relationships:

- Nodes represent **Doctors** and **Patients**.
- Relationships like TREATED connect doctors and patients.
- Enables complex queries, e.g., retrieving all patients treated by a specific doctor.

3. Key-Value Model (Redis):

Redis provided quick access to frequently accessed data:

- Storing appointment details for fast lookups.
 - Tracking doctor visit counts for analytics.
 - Key-value pairs allow rapid retrieval and real-time updates.
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